

Technical Document #276: Natural Language Processing - Comprehensive Overview

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Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Coreference resolution identifies all expressions that refer to the same entity in a text. It is essential for understanding document-level coherence.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Text summarization generates a concise summary of a longer text. Extractive summarization selects important sentences while abstractive summarization generates new text.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Tokenization is the process of breaking text into individual units called tokens.
- Sentiment analysis determines the emotional tone behind a body of text.
- Question answering systems extract answers from a given context.